

Asymmetric Capital Buffer Control: Empirical Companion

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Abstract

This note is the empirical companion to *Asymmetric Capital Buffer Control* (PROOFS_v2); numbered results cited here — Theorem 1, Proposition 11, Remark 1 — refer to that document. It states what the data say about the variance-asymmetry parameter λ_V , against the null of a risk-neutral bank facing the same regulation. Every figure below is regenerated from the scripts named in Section 6; none is carried from an earlier draft. The primary analysis is the pre-registered design, reported exactly as run, including its attrition; a secondary analysis with an alternative reference construction is specified in Section 4 with results deliberately pending its own validation.

1 What is being estimated, and how

Theorem 1 makes λ_V^4 the ratio of conditional variances of the log capital-buffer state deep in deficit to deep in surplus. Two facts about estimating it, both established by simulation (`verify_egarch_recovery.py`), fix the design.

First, θ and λ_V are not jointly identified from routine-sized excursions. With abundant data ($T = 5000$) and the state ranging over $[-1.9, 1.4]$ against $\theta = 1$, the profile log-likelihood in θ is flat to within about one point from $\theta = 0.7$ to $\theta = 5$ while the implied λ_V runs from 1.37 to 6.17: many (c, θ) pairs are close to observationally equivalent, so θ must come from outside the estimation. The practical consequence adopted here is an estimator that does not use θ at all: $\hat{\lambda}$ is the fourth root of the ratio of within-regime innovation variances, each from a within-regime AR(1) residual, so only the regime classification and the variance ratio enter.

Second, conditional on that choice, recovery is clean. At the null (true $\lambda = 1$, $T = 300$, twenty independent replications) the mean bias is +0.085 with standard deviation 0.119, and the bias stays small and one-sided across a (ϕ, T) grid from $\phi = 0.5$ to 0.95 and $T = 150$ to 800 — shrinking to +0.018 at $(\phi, T) = (0.95, 800)$. There is no free trend parameter whose mis-estimation could manufacture asymmetry, which is the failure mode that produced $\hat{\lambda} \approx 2$ at a true null in a sibling project's estimator. The small positive null bias means the estimator, if anything, overstates asymmetry slightly — so it works against, not for, the direction of the finding reported below.

2 The null: what a risk-neutral bank would already show

The comparison value is not $\lambda_V = 1$. A risk-neutral bank ($\lambda_S = 1$) facing the regulatory cap already generates asymmetric conditional variance, because the cap tightens exposure as the buffer erodes; the honest null is therefore λ_V computed from the solved model itself. Two routes exist and only one survives.

The *definitional* route — Theorem 1’s ratio evaluated at the saturation limits — is a constant of cap geometry, $\lambda_V = (\kappa_1/\kappa_3)^{1/2} = 2.599$ at the baseline, with no dependence on λ_S or any preference or cost parameter (Proposition 11). It cannot serve as a benchmark for data that never reach the saturation limits, and it carries no information about λ_S .

The *estimator* route applies the same θ -free estimator to paths simulated from the solved model. Computed on the convergence-certified solves (Remark 1) at $\lambda_S \in \{1.00, 1.10, 1.25, 1.50, 1.75, 2.00\}$, under three conventions:

Convention	λ_S					
	1.00	1.10	1.25	1.50	1.75	2.00
(a) RWA~L, split at stationary mean	1.3320	1.3269	1.2892	1.2724	1.2477	1.2347
(b) RWA~A, split at stationary mean	1.4762	1.4764	1.4259	1.4038	1.3723	1.3545
(a') RWA~L, split fixed at R	1.3100	1.3423	1.3590	1.3721	1.3967	1.4056

Three properties of this table matter. The $\lambda_S = 1$ column is the risk-neutral null under each convention: 1.31–1.48. The *direction* in which λ_V moves with λ_S is convention-dependent: both moving-threshold conventions decrease (slopes -0.10 and -0.13 per unit λ_S), while the fixed-threshold convention increases ($+0.09$); the likely mechanism is that under a moving threshold the split point itself shifts with λ_S (from 1.30 to 1.43 across the grid), confounding the λ_S effect with a changing comparison point. The panel estimator of Section 3 classifies regimes against a moving, history-based threshold, so rows (a)–(b) are the relevant comparison for it, and under them λ_V *falls* as asymmetry rises. Finally, the smallest value anywhere in the table is 1.2347: no admissible λ_S in the solved range produces λ_V below that, under any convention.

3 Primary analysis: the pre-registered design, as run

Run on 11 August 2026 against the live IMF FSI panel under the stated universe rule (`panel_universe.py`: every sovereign or dependent-territory area the source reports, no history pre-screen; 152 areas returned, 151 sovereign, one regional aggregate excluded by name). Of the 151, 114 carry enough history to attempt estimation and 37 are too short. The estimator produces a usable $\hat{\lambda}$ for **33 countries**: median **1.038**, mean 1.132, interquartile range [0.963, 1.288], fraction above one 0.61. Ten of the 33 sit at or above the lowest model value; thirteen sit below one. One estimate (Equatorial Guinea) carries a within-regime AR coefficient of -3.9 and should be treated as unreliable regardless of its regime counts.

The headline is robust to the universe correction: the previous, unexplained 77-country list gave median 1.0395 on $n = 26$; the full sovereign universe, roughly twice the size, gives 1.0379 on $n = 33$. Doubling the universe moved the median by 0.002.

3.1 The attrition, stated

Of the 114 attempted, 81 do not yield an estimate, and the two mechanisms must be reported separately because only one is curable by threshold choice. **Structural** exclusions — zero quarters in one regime, which no per-regime minimum can repair — account for 24 countries, of which eighteen are advanced or EU economies: France (39 surplus quarters, 0 deficit), the United Kingdom (44/0), Belgium (49/0), Sweden (39/0), Austria (51/0), Ireland (48/0), Israel (64/0), the Netherlands (56/0), Norway (52/0), and the remainder in central and northern Europe. **Near-miss**

exclusions — a nonzero smaller regime short of the per-regime minimum of 15 — include Switzerland, Spain and Denmark at 2 deficit quarters each, Luxembourg at 1, Greece at 3, Portugal at 6, and the United States at 14.

The sharpened form of the selection finding is therefore: *under a one-sided expanding-mean reference, the deficit regime is unobservable in every advanced banking system in the post-Basel III sample*. Not thinly observed — unobservable. A capital ratio that trends upward from the start of its reported history never falls below its own expanding mean, and the advanced economies are precisely the trending ones. An advanced-economy-only panel under this construction would have $n = 4$ (Canada, Estonia, Macao, San Marino). The measurable population is selected on the dynamics of the variable being measured, and the population of greatest interest for the theory is the one selected out. Section 4 exists for exactly this reason.

3.2 Reading the result

The comparison must be made on the measured scale. The estimator is biased toward one, and the bias does not vanish with sample size: a process with true $\lambda = 1.30$ measures 1.16 at $T = 60$ quarters and still 1.14 at $T = 20,000$, because observations near the classification boundary are assigned to a regime while their innovations reflect a mixture of both. It is a different estimand, not a small-sample artefact. The model’s population value (Section 2) is therefore not the right benchmark for the panel median; the right benchmark is what the panel’s own estimator returns when applied to the model (`results/lambda_RN_measured.py`: the solved model simulated at quarterly frequency over 60-quarter windows, estimated by the identical code path, 400 paths per parameter value).

On that scale, the model class spans median measured values of **1.334** (risk-neutral, convention (a)) down to **1.152** (the most asymmetric admissible parameter, $\lambda_S = 2$), with convention (b) slightly higher throughout, and a per-path sampling standard deviation of 0.16–0.20.

Against this the panel median of 1.038 sits below every cell of the grid: 0.30 below the risk-neutral benchmark (roughly six to seven standard errors of a median of 33) and 0.11 below even the most asymmetric admissible model (roughly two to three standard errors). The direction is unambiguous under both conventions and every admissible λ_S : *measured deficit-state volatility in the usable panel is lower than a preference-free regulated bank would be measured at*. The magnitude, however, is modest once stated on the measured scale, and no claim is made here that a preference parameter is identified from it: 13.2% of individual simulated model banks measure at or below 1.038 over sixty quarters, so any single country at that level is unremarkable and the evidence rests entirely on the median of 33. The population value of the gap is roughly 0.20 if the attenuation mapping is inverted, but that inversion is model-dependent and is not used for any claim.

Two further cautions. First, the measured-scale model values are far flatter in λ_S (1.334 to 1.152 across the whole admissible range) than the population values, so the data compress exactly the variation that would be needed to read λ_S back from a variance ratio — consistent with the identification analysis in Section 2, and one more reason the committed thresholds, not the variance ratio, are the identifying moments if identification is ever attempted. Second, the usable panel is the mean-reverting, largely emerging-market subsample that the reference construction itself selected; the undershoot is a fact about that population, generalised to no other.

4 Secondary analysis: alternative reference (specified; results pending validation)

The attrition of Section ?? motivates a second reference construction, recorded here *before* its results are computed, for four reasons: the theory’s reference is a threshold, not a country’s own trend, so “below your own past” is not the estimand Theorem 1 defines; the surviving sample is selected on behaviour plausibly correlated with the quantity measured; the excluded countries are exactly the deep-banking-system economies the theory most concerns; and the model null of Section 2 is computed from a stationary distribution, so the comparison currently holds only on the subsample where stationarity approximately obtains.

Construction. Per country, the reference is the full-sample mean of the log ratio, $R_i = \text{mean}(\text{logratio}_{i,\cdot})$, constant over time; classification and the θ -free estimator are otherwise unchanged. A trending series lies below its full-sample mean early and above it late, so both regimes exist for every country and the 13 zero-deficit countries re-enter.

Known costs, stated in advance. The construction uses look-ahead — the reference at date t involves data after t — and for a trending series the regime label partially proxies for calendar time, so $\hat{\lambda}$ may pick up secular changes in volatility rather than state-dependence. Both effects are testable in simulation, and the design is therefore gated exactly as the primary was: it runs on simulated paths first — recovery against the true- z benchmark, a clean null with no manufactured asymmetry, and additionally a *trended* null (upward-drifting ratio, symmetric shocks), which is the case the look-ahead concern targets — and touches real data only if it passes all three. No bandwidth or window parameter is introduced, so there is no tuning margin between this specification and its result.

Interpretation rule, fixed now. If the secondary agrees with the primary (median well below the null range), the finding generalises beyond the mean-reverting subsample. If it disagrees, the primary result is evidence about the reference construction, not about banks, and neither number should be quoted without the other.

5 What this note does not establish

The calibration behind the null is the project’s stylised parameter set, not fitted to bank data. The pre-committed ϕ^- test, with its fixed treatment of the nine ROE $< g$ countries, remains unexecuted, as does the bank-level feasibility check (`empirical_design_exclusions`). The RWA~A versus RWA~L choice is settled by the definitional argument that Basel’s risk weights are asset-side, but has not been checked against the IMF’s construction of FSKRC_PT; the primary reading does not depend on it, since the median sits below the null under both. And whether the solved model’s constructed value function is the value function — the verification theorem — remains open (PROOFS_v2, “What is not established”), so the null values here are properties of the certified numerical solution.

6 Reproduction

Panel: `03_empirical/design/expanding_mean_panel.py` (live IMF FSI fetch; requires network and `statsmodels`); run 11 August 2026, output reproduced in Section 3. Null curve: `02_numerical/run_lambda_sw` at `max_iter` ≥ 2000 , then `run_lambda_RN_curve.sh` against `lambda_RN_panelconv.m`; the certified solves and the resulting `lambda_V_curve.csv` are in `03_empirical/results/`. Estimator proper-ties: `01_theory/verify_egarch_recovery.py` (Parts A, B, E quoted in Section 1). The secondary

analysis of Section 4 has, deliberately, no results file yet; its validation harness extends the same script.